

Abhinav Singh

India • +91 9354726744 • abhinav2006singh@gmail.com • [LinkedIn](#) • [GitHub](#)

EDUCATION

Bennett University, Greater Noida, UP
B.Tech – Computer Science & Engineering

Expected Jun 2028

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Solidity, CSS

Blockchain / Web3: Solidity, ERC-721, Hardhat, ethers.js, IPFS, Circom, Anchor (Solana)

Frontend / Backend: React.js, Next.js, Node.js, Tailwind CSS, FastAPI

Databases: PostgreSQL, MySQL, MongoDB

Frameworks / Libraries: OpenZeppelin, wagmi, RainbowKit, Prisma, snarkjs, PyTorch, Scikit-Learn

PROJECT EXPERIENCE

ZK-Whistleblower – Anonymous Blockchain Reporting Platform ([GitHub](#) | [Live Demo](#)) Feb 2026 – Present

- Built a full-stack anonymous whistleblowing platform using ZK-SNARKs (Groth16, Circom 2), allowing organisation insiders to prove membership — via Poseidon-hashed Merkle trees and nullifier hashes — without revealing their identity.
- Developed Solidity smart contracts (Hardhat 3 + OpenZeppelin) for on-chain ZK proof verification, Merkle root management, and tamper-proof IPFS-CID evidence storage.
- Architected a Next.js 15 monorepo (Admin + Reporter apps) with wagmi v2 + RainbowKit and client-side Groth16 proof generation in the browser via snarkjs.
- Wrote 13 tests covering valid proofs, nullifier replay attacks, revoked roots, and fake proofs — all using real ZK proof generation with zero mocks.

ERC-721 NFT Smart Contract – Scalable NFT System ([GitHub](#)) Mar 2026 – Apr 2026

- Implemented a production-grade ERC-721 contract with owner-minted and public fee-based minting, max supply cap, and auto-incrementing token IDs using OpenZeppelin.
- Added security features: ReentrancyGuard for payable flows, Pausable controls, and owner access control with a public mint toggle; integrated IPFS-based per-token metadata URI storage.
- Deployed and verified on Ethereum Sepolia testnet with interaction scripts for ownerOf() and tokenURI() reads.

Sol-Uptime – Decentralised Uptime Monitor on Solana ([GitHub](#)) Aug 2025 – Nov 2025

- Collaborated on a decentralised ping/uptime monitoring system built as a Solana on-chain program using the Anchor framework in Rust, deployed to Solana Devnet.
- Contributed to the Next.js frontend integrating @solana/web3.js wallet UI and Anchor-generated client code, demonstrating multi-chain competency across EVM and non-EVM ecosystems.

Authentiq – AI-Powered Deepfake & Synthetic Content Detector ([GitHub](#)) Feb 2026 – Apr 2026

- Collaborated on a two-tier deepfake detection platform: Next.js 16 frontend communicating with a FastAPI backend serving live PyTorch/Torchvision inference models for images and Scikit-Learn classifiers for AI-generated text.
- Integrated a unified dev startup script booting both services concurrently; supports images (.jpg, .png, .webp) and documents (.pdf, .docx, .txt, .csv) via python-multipart upload handling.

CERTIFICATIONS

- NPTEL – [Blockchain and Its Applications](#)
- NPTEL – [Data Structures and Algorithms Design](#)
- Coursera – [The Bits and Bytes of Computer Networking](#)
- Coursera – [Operating Systems](#)

ACHIEVEMENT

NPTEL Blockchain and Its Applications — Elite + Silver Medal, Top 5% nationally.